INCH-POUND

MIL-A-82772(OS)
29 December 1989
SUPERSEDING
WS 21903
25 October 1983

MILITARY SPECIFICATION

ASSEMBLY, AFT CLOSURE, IN-PROCESS REQUIREMENTS FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the in-process requirements for one type of aft closure assembly, referred to herein as the assembly (see 6.1). The in-process requirements herein cover assembly characteristics which must be verified prior to assembly and fabrication of the end product.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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STANDARDS

MILITARY

MIL-STD-453 Inspection, Radiographic

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 **Non-Government publications.** The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 740 Methyl Ethyl Ketone (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia Absorbent Gauze

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 In-process. The assembly shall conform to the requirements specified herein.

3.2 Surface condition. The surface condition of the assembly shall not exceed the permissible extent specified in table I.

TABLE I. Surface defect conditions.

Component	Defect	Permissible Condition Extent
Throat inlet, closure insulation and throat support	Blemishes (see 6.3.1)	Shall not exceed one square inch, 2.0 inches in length or 0.03 inch in depth. No two greater than 0.5 square inch or 0.03 inch in depth shall be closer together than two inches. No more than three per part.
	Wrinkled areas (see 6.3.4)	None allowed.
	Delaminations (see 6.3.3)	None allowed.
	Cracks (see 6.3.2)	None allowed.
Aft closure shell "O" ring surfaces, lockwire groove, threads, major O.D., and throat insert	Blemishes	None allowed.
Other coated surfaces of aft closure shell	Blemishes	None deeper than .002 inch.

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3.3 Subsurface condition. The subsurface condition of the assembly shall not exceed the permissible condition extent specified in table II.

TABLE II. Subsurface defect limits.

Defect	Permissible Extent
a. Cracks (see 6.3.2)	None allowed.
b. Low density line (see 6.3.5)	None greater than 0.050 inch wide. No more than 5 per part. (A low density line between the same two fabric plies which extends over part circumference is considered a single line.)
c. Low density areas (see 6.3.6)	Any single area less than 0.100 inch in length and 0.100 inch in width. No more than one in any 1 square inch area. No more than 5 per part.
d. High density inclusions (see 6.3.7)	Any single inclusion shall be less than 0.050 inch in length or width in the plane of the radiograph. Total surface area of inclusions shall be less than 0.025 square inch in any square inch.
e. Delaminations (see 6.3.3)	None allowed.

3.4 Interface bonds. Separations, unbonds and voids in the interface bonds shall not exceed the permissible condition extent specified in table III (see figure 1).

3.5 Workmanship. The assembly shall be free from surface contaminants, grease, oil, foreign material, or other defects which could render the part unsuitable for the intended use.

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TABLE III. Interface bond defect criteria.

Component	Permissible Extent (As Measured on Any Single Tangential Radiograph)
1. Closure shell to: a. Closure insulation	No single separation or void length greater than 0.75 inch. Accumulative separation or voids less than 20 percent of the bond length.
b. Throat support	No single separation or void length greater than 0.25 inch. Accumulative separation or voids less than 20 percent of the bond length.
2. Throat inlet to: 1/ a. Closure insulation b. Throat insert c. Throat support	No single separation or void length greater than 0.50 inch. Accumulative separation or voids less than 20 percent of the bond length.
3. Throat insert to: a. Throat support	No single separation or void length greater than 0.25 inch. Accumulative separation or voids less than 20 percent of the bond length.
4. Throat support to: a. Closure insulation	No single separation or void greater than 0.08 inch. Accumulative separation or voids less than 50 percent of the bond length.

1/ The throat inlet to closure insulation, throat insert and throat support bond line is considered as one continuous bond line and the void and separation criteria apply as such.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the inspections specified in table IV.

TABLE IV. Quality conformance inspection.

Inspection	Requirement	Inspection Method
Visual examination	3.2 & 3.5	4.4.1
Radiographic inspection	3.3 & 3.4	4.4.2

4.3 Classification of characteristics. The characteristics verified by the inspections herein are classified as Major or Minor in accordance with WR-43. Inspections that verify major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the inspection. The complete visual examination herein is not annotated with a classification code or serial number and is classified Minor. The portion of the visual examination to which a major classification is applicable is enclosed in brackets.

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4.4 **Methods of inspection.** The following test inspection methods and procedures shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

4.4.1 **Visual examination.** Each assembly including aft closure shell "O" ring surfaces, lockwire groove, threads, major O.D., and throat insert (M101) shall be visually examined to determine conformance to 3.2 and 3.5. Noncompliance with any specified requirement shall constitute cause for rejection. The following methyl ethyl ketone (MEK) solvent wipe method shall be used as an examination aid in determining conformance to surface crack and delamination requirements specified in 3.2.

a. Wet the surface with MEK conforming to ASTM D 740.

b. Wipe off excess MEK with type VIII absorbent gauze conforming to The United States Pharmacopeia.

c. Allow part to dry in air. Defects will appear as wetted lines or areas when surrounding area appears dry.

4.4.2 **Radiographic inspection (M102).** Each assembly shall be radiographically inspected in accordance with MIL-STD-453 and figure 2 to determine the subsurface condition as specified in 3.3 and 3.4. The finished radiograph shall demonstrate 1.5 percent tangential sensitivity. Noncompliance with any specified requirement shall constitute cause for rejection.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 **Intended use.** The aft closure assembly specified herein is intended for use as a component in the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

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6.3 **Definitions**

6.3.1 **Blemishes.** Any superficial imperfection or marks or deformities such as scratches, crazing, gouges, cuts, or stains.

6.3.2 **Cracks.** A separation which extends across fabric plies.

6.3.3 **Delaminations.** A separation between fabric plies which is parallel to the fabric plies.

6.3.4 **Wrinkled areas.** Distortion in fabric orientation normal to the plane of the fabric.

6.3.5 **Low density line.** A low density area between fabric plies which is parallel to the fabric plies.

6.3.6 **Low density area.** A low density area which has appreciable width as measured perpendicular to the plies and is not necessarily oriented in any specific manner with respect to the plies.

6.3.7 **High density inclusion.** Object that has higher density than that of the surrounding material.

6.4 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1340-N781)

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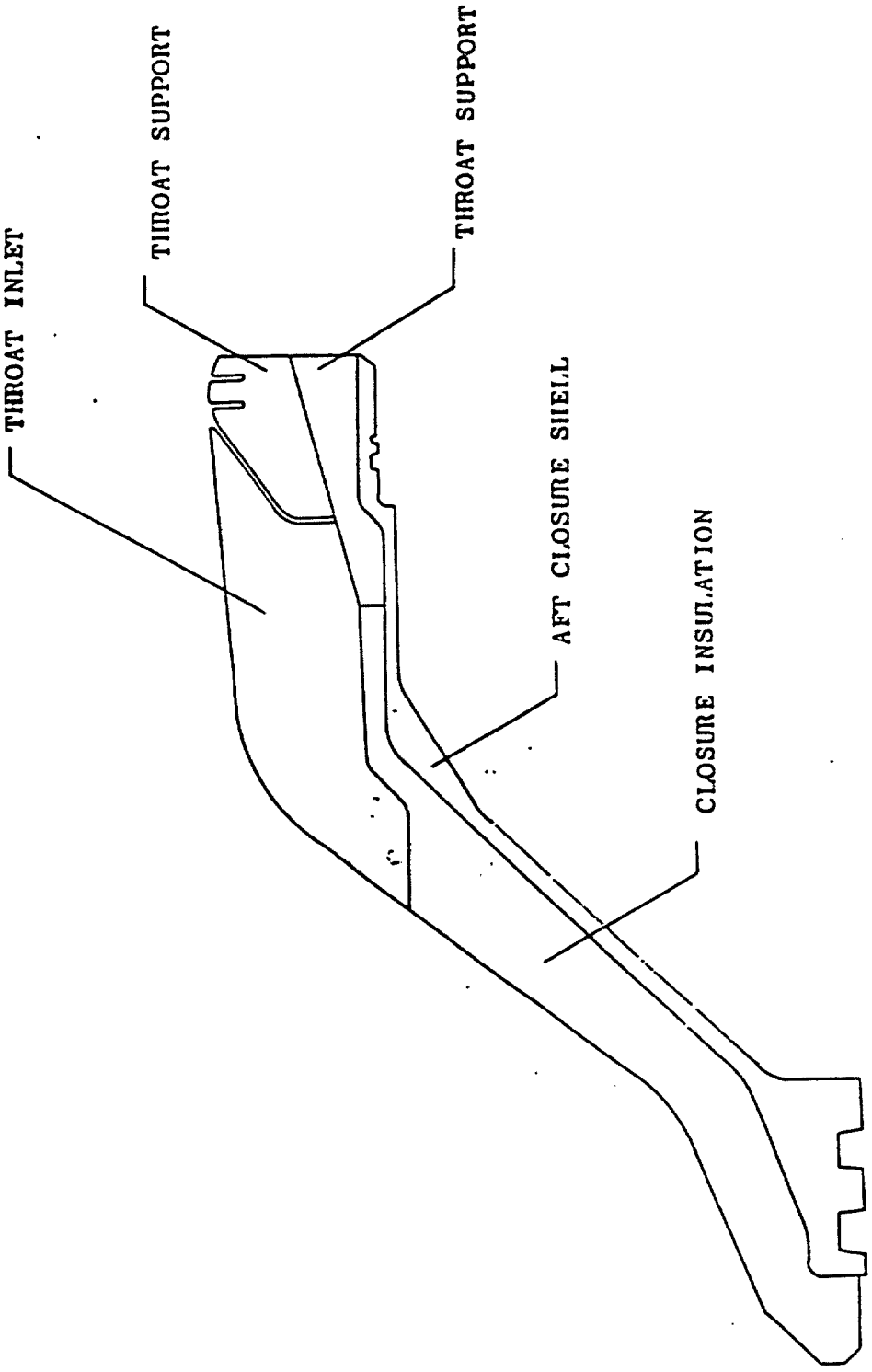


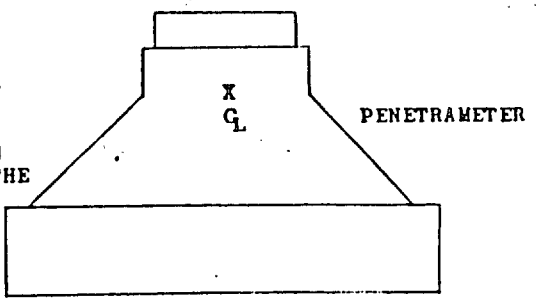
FIGURE 1. Aft closure assembly.

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VIEW	SOURCE	KVP	Ma	EXP TIME	FFD	FILM	Pb SCREENS	PENETRAMETER
1	LINAC	4MEV	-	240R	25'	14X17 M&R	.02-.02 F&B	.018 Fe WIRE
ALT	LINAC	8MEV	-	200R	25'	14X17 M&R	.02-.02 F&B	.018 Fe WIRE

NOTES

1. DEGREES SHALL BE MEASURED AFT LOOKING FORWARD IN A CLOCKWISE DIRECTION TO PRODUCE TANGENTIALS EVERY 60°.
2. IDENTIFY FILM, USING LEAD NUMBERS, WITH DEGREES AT TANGENT POINT AND THE LAST THREE DIGITS OF THE PART SERIAL NUMBER.
3. BEAM CENTERLINE SHALL BE LOCATED 3.00"+/-.100" FORWARD OF AFT END OF PART.



VIEW 1

FIGURE 2. Radiographic inspection requirements.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-82772	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
ASSEMBLY, AFT CLOSURE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

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